Hunmin Yang

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Senior AI Researcher @ ADD
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Research Interest

- **Trustworthy Machine Learning**
  - My research interests broadly lie in machine learning and computer vision, with a recent focus on generative AI, particularly multimodal large language models (LLMs) and diffusion models. My previous work includes leveraging generative models for transferable adversarial attacks and building robust AI systems that can operate reliably in diverse environments. I hope to continue making a positive impact on society by enhancing the safety and trustworthiness of AI.

Experience

- **Agency for Defense Development (ADD)** — Daejeon, Korea
  Senior AI Researcher — *Jan 2020 - Present*
  - **D-CAM**: Adversarial attack & defense techniques for robust AI
  - **D-GEN**: Synthetic data generation framework for training AI
- **Agency for Defense Development (ADD)** — Daejeon, Korea
  AI Researcher — *May 2017 - Dec 2019*
  - **D-NET**: Large-scale AI inference platform with Hadoop-Spark
- **Agency for Defense Development (ADD)** — Daejeon, Korea
  Specialized Research Staff (Military Service) — *Feb 2014 - May 2017*

Education

- **Korea Advance Institute of Science and Technology (KAIST)** — Daejeon, Korea
  PhD in Mechanical Engineering — *Sep 2021 - Feb 2026*
  - Research Area: Adversarial Machine Learning
  - Advisor: Kuk-Jin Yoon
- **Korea Advance Institute of Science and Technology (KAIST)** — Daejeon, Korea
  MS in Mechanical Engineering — *Feb 2012 - Feb 2014*
  - Research Area: 3D Sound Perception
  - Advisor: Youngjin Park
- **Royal Melbourne Institute of Technology (RMIT)** — Melbourne, Australia
  Exchange Student (High Distinction) — *Feb 2011 - Aug 2011*
- **Korea Advance Institute of Science and Technology (KAIST)** — Daejeon, Korea
  BS in Mechanical Engineering (Magna Cum Laude) — *Feb 2007 - Feb 2012*

Publications

- Se-Yoon Oh, **Hunmin Yang**. Physical Adversarial Attacks and Defenses on Vehicle Detection Models. In *International Conference on Control, Automation and Systems (ICCAS)*, 2025.
- Jongoh Jeong, **Hunmin Yang**, Kuk-Jin Yoon. Boosting Adversarial Transferability with a Generative Model Perspective. In *CVPR Workshop on Generative Models for Computer Vision (GMCV)*, 2025.
- Junhyeong Cho, Kim Youwang, **Hunmin Yang**, Tae-Hyun Oh. Robust 3D Shape Reconstruction in Zero-Shot from a Single Image in the Wild. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- **Hunmin Yang**, Jongoh Jeong, Kuk-Jin Yoon. Prompt-Driven Contrastive Learning for Transferable Adversarial Attacks. In *European Conference on Computer Vision (ECCV)*, 2024. (**Oral, top 2.3%**)
- Junhyeong Cho, Kim Youwang, **Hunmin Yang**, Tae-Hyun Oh. Object-Centric Domain Randomization for 3D Shape Reconstruction in the Wild. In *CVPR Workshop on Foundation Models (WFM)*, 2024.
- **Hunmin Yang***, Jongoh Jeong*, Kuk-Jin Yoon. FACL-Attack: Frequency-Aware Contrastive Learning for Transferable Adversarial Attacks. In *Association for the Advancement of Artificial Intelligence (AAAI)*, 2024.

- Junhyeong Cho, Gilhyun Nam, Sungyeon Kim, **Hunmin Yang**, Suha Kwak. PromptStyler: Prompt-driven Style Generation for Source-free Domain Generalization. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- Naufal Suryanto, Yongsu Kim, Harashta Tatimma Larasati, Hyoeun Kang, Thi-Thu-Huong Le, Yoonyoung Hong, **Hunmin Yang**, Se-Yoon Oh, Howon Kim. ACTIVE: Towards Highly Transferable 3D Physical Camouflage for Universal and Robust Vehicle Evasion. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- **Hunmin Yang**, Se-Yoon Oh, Junhyeong Cho. Synthetic Image Generation for Deep Neural Networks. In *NVIDIA GPU Technology Conference (GTC)*, 2023. (**Spotlight**)
- Naufal Suryanto, Yongsu Kim, Hyoeun Kang, Harashta Tatimma Larasati, Youngyeo Yun, Thi-Thu-Huong Le, **Hunmin Yang**, Se-Yoon Oh, Howon Kim. DTA: Physical Camouflage Attacks using Differentiable Transformation Network. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Jeonghun Kim, Kyungmin Lee, Hyeongkeun Lee, **Hunmin Yang**, Se-Yoon Oh. Camouflaged Adversarial Attack on Object Detector . In *21th International Conference on Control, Automation and Systems (ICCAS)*, 2021.
- **Hunmin Yang**, Se-Yoon Oh, Taewon Kim, Ki-Jung Ryu. D-GEN: A Deep Learning Data Generation Framework For Artificial Intelligence. In *NVIDIA GPU Technology Conference (GTC)*, 2020.
- Kyungmin Lee, **Hunmin Yang**, Se-Yoon Oh. Adversarial Training on Joint Energy Based Model for Robust Classification and Out-of-Distribution Detection. In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
- Eunchong Kim, Kanghyun Park, **Hunmin Yang**, Se-Yoon Oh. Training Deep Neural Networks with Synthetic Data for Off-Road Vehicle Detection. In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
- Hyeongkeun Lee, Kyungmin Lee, **Hunmin Yang**, Se-Yoon Oh. Applying FastPhotoStyle to Synthetic Data for Military Vehicle Detection. In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
- Kanghyun Park, Hyeongkeun Lee, **Hunmin Yang**, Se-Yoon Oh. Improving Instance Segmentation using Synthetic Data with Artificial Distractors. In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
- **Hunmin Yang**, Se-Yoon Oh, Ki-Jung Ryu. Accelerating Distributed Deep Learning Inference on multi-GPU with Hadoop-Spark. In *NVIDIA GPU Technology Conference (GTC)*, 2019. (**Oral**)
- **Hunmin Yang**, Se-Yoon Oh, Ki-Jung Ryu. Scalable Distributed Deep Learning Inference on Multi-GPU with Hadoop-Spark. In *NVIDIA GPU Technology Conference (GTC)*, 2019.
- Se-Yoon Oh, **Hunmin Yang**, Ki-Jung Ryu. Optimal Distributed Inference on Multi-GPU Processing System. In *NVIDIA GPU Technology Conference (GTC)*, 2019.
- Se-Yoon Oh, **Hunmin Yang**, Ki-Jung Ryu. Optimal Experimental Design Approach for Machine Learning Process. In *International Conference on Control, Automation and Systems (ICCAS)*, 2017.

Patents

Machine Learning & Synthetic Image Generation

- **Hunmin Yang**, Se-Yoon Oh. Training data generation method and apparatus for deep learning model. kr 10-2613781, 2023.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Apparatus and method for deep learning based on mixing virtual and real data. kr 10-2198088, 2020.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Apparatus and method for learning machine learning models based on virtual data. kr 10-2086351, 2020.
- **Hunmin Yang**, Se-Yoon Oh, Seongbaek Jo. Apparatus and method for enhancing learning capability for machine learning. kr 10-2053202, 2019.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus of improving self-supervised learning performance utilizing synthesized data. kr 10-2032519, 2019.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and Apparatus of adding artificial object for improving performance in detecting object. kr 10-1972095, 2019.
- **Hunmin Yang**, Se-Yoon Oh, Seongbaek Jo. Apparatus and method for generating learning image in game engine-based machine learning. kr 10-1947650, 2019.

AI Security & Adversarial Robustness

- Se-Yoon Oh, **Hunmin Yang**, Hyeongkeun Lee, Kyungmin Lee, Jeonghun Kim. Method and Apparatus for optimizing adversarial patch, computer-readable storage medium and computer program. kr 10-2445215, 2022.
- Hyeongkeun Lee, Jeonghun Kim, Kyungmin Lee, **Hunmin Yang**, Se-Yoon Oh. Method and Apparatus for optimizing adversarial patch, computer-readable storage medium and computer program. kr 10-2414146, 2022.
- Jeonghun Kim, Se-Yoon Oh, Hyeongkeun Lee, Kyungmin Lee, **Hunmin Yang**. Apparatus and method for optimizing adversarial patch based on natural pattern for stealthiness against human vision system. kr 10-2380154, 2022.
- Hyeongkeun Lee, **Hunmin Yang**, Jeonghun Kim, Kyungmin Lee, Se-Yoon Oh. Method, apparatus computer-readable storage medium and computer program for determining adversarial patch position. kr 10-2360070, 2022.

Big Data & Database

- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus of building NoSQL database for signal processing. kr 10-2002360, 2019.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus of building inverse index DB for high speed searching of moving picture object. kr 10-2014267, 2019.

Honors and Awards

- **Defense Science Award** — Aug 2025
  From the Chief Research Director of ADD
  - Adversarial Attack Countermeasures for Robust AI-based Image Analysis Systems
- **Selected as Oral Presentation (top 2.3%)** — Oct 2024
  In European Conference on Computer Vision (ECCV)
  - Prompt-Driven Contrastive Learning for Transferable Adversarial Attacks
- **National Grant for Defense Research and Development** — Dec 2021
  From the Chief Director of DAPA
  - Synthetic Data Generation for Defense AI
- **Defense Science Award** — Aug 2019
  From the Chief Research Director of ADD
  - Improving Distributed Multi-GPU Computing for Large-scale Video Analytics
- **High Achievement Award** — Aug 2018
  From the Chief Research Director of ADD
  - Big Data Platform Development and Synthetic Data Generation
- **Excellent Paper Award** — Mar 2013
  From the Korea Society for Noise and Vibration Engineering (KSNVE)
  - Sweet Spot Analysis of Linear Array System by Geometrical Approach
- **Scholarship for Academic Excellence** — Jun 2007
  From the Korea Human Resource Development Scholarship Association
  - Outstanding Academic Performance
- **Scholarship for Academic Excellence** — 2007-2014
  From the Korean Government
  - Tuition free for all semesters in KAIST (BS & MS)

Professional Services

- **Academic Reviewer**
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
  - IEEE/CVF International Conference on Computer Vision (ICCV)
  - European Conference on Computer Vision (ECCV)
  - Neural Information Processing Systems (NeurIPS)
  - AAAI Conference on Artificial Intelligence (AAAI)
- **Technology Transfer**
  - Synthetic data generation for training AI models → SI Analytics, Jcorp System, JEIOS
  - Physical adversarial camouflage generation for attacking AI models → SmartM2M
  - Big data platform for large-scale AI model inference → XIIlab